

Shoulder pain

by ANN MJ COOLS*

When I dislocated my shoulder back in 2005 and (went through) rehab throughout 2006, I was told by Dr Andrews 'you're always going to have to stay on top of that shoulder. You're always going to have to do a little bit extra to keep it at the level you want to keep it at'. In a lot of ways it was the best thing that ever happened to me because I started doing things that I'd never done before. So I learned so much about my shoulder and how to manage my shoulder.

Drew Brees, NFL quarterback

Shoulder pain and dysfunction are a significant health issue, in the general population as well as in an athletic population. Whilst acute shoulder injuries occur mainly in collision sports or as a result of a sports accident like falling in skiing or biking, athletes performing overhead sports such as tennis, baseball, swimming and gymnastics are also prone to chronic shoulder pain, due to the high demands on their shoulders during their sports performance. In this chapter, the following topics regarding assessment and treatment of shoulder pain are discussed:

- functional anatomy and biomechanics
- overview of most common, less common, and not-to-miss causes of shoulder pain
- clinical examination: from history and basic clinical exam to special tests and outcome scores
- investigation
- general treatment and specific rehabilitation guidelines for the most common shoulder injuries in athletes
- less common causes of shoulder pain: overview of treatment guidelines
- special considerations for the overhead athlete: the thrower's program, sport-specific kinetic chain principles and return to play after injury.

FUNCTIONAL ANATOMY AND BIOMECHANICS

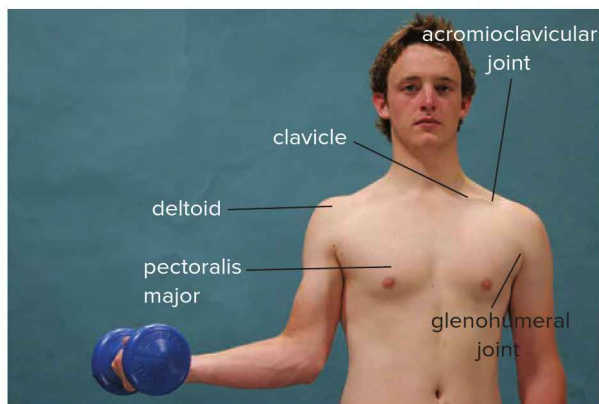
The glenohumeral joint is an inherently unstable shallow ball and socket joint, often described as the equivalent of

a golf ball (head of humerus) on a tee (glenoid). In fact, the relationship between the humeral head and the glenoid cavity more accurately parallels a sea lion balancing a ball on its nose. Thus, effective shoulder function and stability requires both static constraints—the glenohumeral ligaments, glenoid labrum and capsule—and dynamic constraints, predominantly the rotator cuff and scapular stabilising muscles (Fig. 24.1).

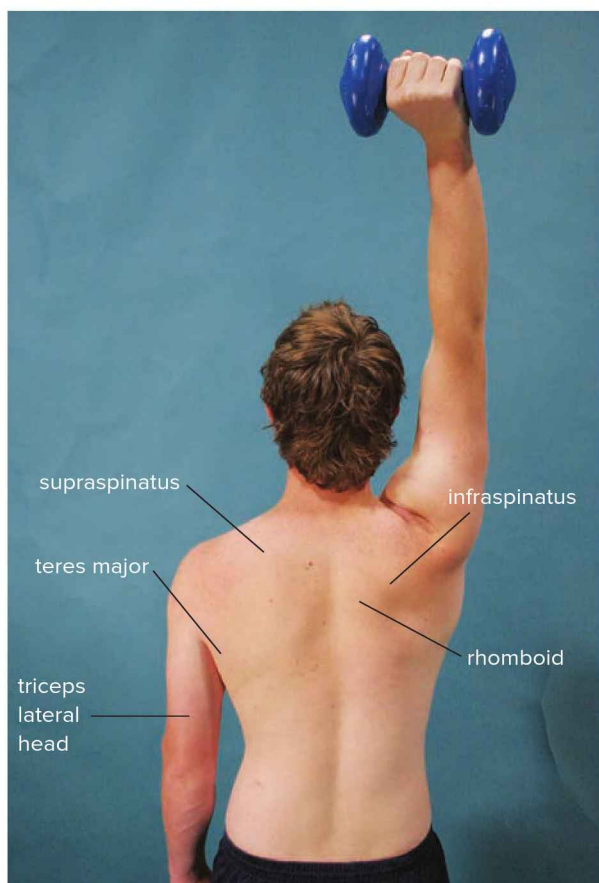
Static stabilisers

In the neutral position, the coracohumeral ligament and the superior glenohumeral ligament control inferior translation of the humeral head. In an abducted position, all parts of the inferior glenohumeral ligament prevent excessive inferior translation. During external rotation in slight elevation (0–45°), the medial glenohumeral ligament prevents anterior translation. The main static stabilisers of the shoulder in the functional position (abducted) are the anterior and posterior bands of the inferior glenohumeral ligament. Together with the axillary pouch, these three parts of the inferior glenohumeral ligament (IGHL) are the thickest of all glenohumeral ligaments (Fig. 24.1d). They are attached to the labrum, which, in turn, attaches directly to the margin of the glenoid fossa. The anterior band of the inferior glenohumeral ligament prevents anterior translation and the posterior band prevents posterior translation of the humeral head. The superior margin of the anterior band of this ligament attaches to the

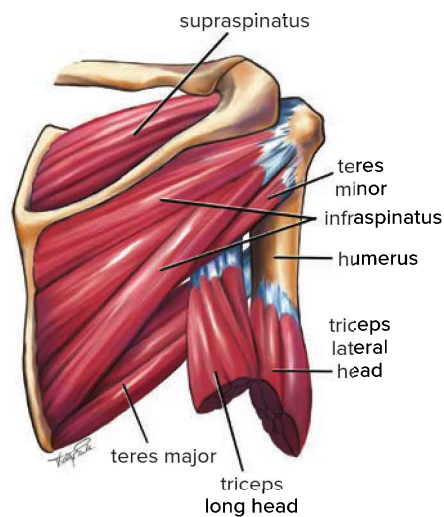
* 4th edition authors: W Ben Kibler, George AC Murrell and Babette Pluim



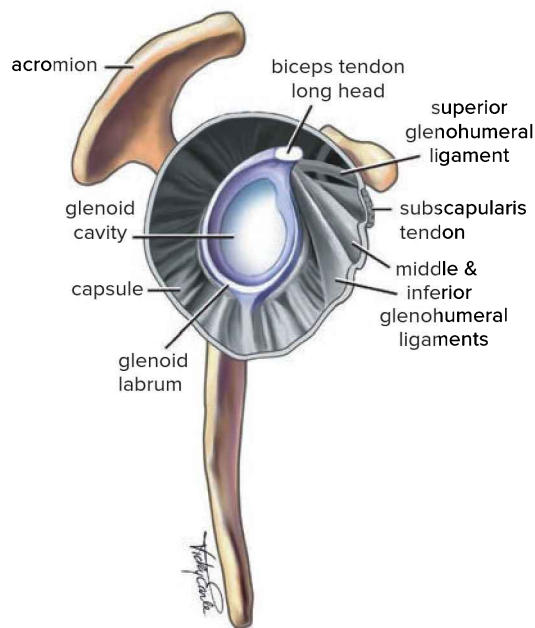
(a)



(b)



(c)



(d)

Figure 24.1 Anatomy of the shoulder region (a) Surface anatomy from the front (b) Surface anatomy from behind (c) Rotator cuff musculature from behind (d) Ligaments and muscles around the glenohumeral joint

glenoid fossa anteriorly at the two o'clock position. When the arm is placed into abduction and external rotation, this broad ligamentous band of the IGHL rotates anteriorly to prevent subluxation of the joint.¹ In addition, the

IGHL seems to have a dual role—it also forms a substantial posterior barrier when the arm is internally rotated in the abducted or forward flexed position. Shoulder stability is also enhanced by the glenoid labrum, a ring of fibrous

tissue attached to the rim of the glenoid, which expands the size and depth of the glenoid cavity. It increases the superior-inferior diameter of the glenoid by 75% and the anterior-posterior diameter by 50%. Moreover, the plasticity of the glenoid allows expanding the size of the glenoid fossa without losing range of motion, and is an attachment site for ligaments and the biceps tendon.

Dynamic stabilisers

The dynamic stabilisers of the glenohumeral joint are the rotator cuff muscles, consisting of the supraspinatus, infraspinatus, subscapularis and teres minor, which act in co-contraction to seat the humeral head in the glenoid (Fig. 24.1c). Since the tendons splay out and interdigitate to form a common, continuous insertion to the humerus, their roles are highly interrelated. The specific roles of the different tendons are debated in literature. The most recent studies attribute similar stabilising roles to infraspinatus and supraspinatus performing external rotation, and avoiding anterior translation of the humeral head, whereas subscapularis is contracting to avoid posterior translation.² In addition, the rotator cuff muscles counteract the action of the deltoid (which elevates the humeral head) by preventing the head of the humerus from moving superiorly when the arm is raised. The key issue in this force couple concept of the rotator cuff, is muscular balance between the different components of the rotator cuff, in particular between the internal and external rotators (ER/IR ratio). Overhead athletes often exhibit stronger internal rotators, with subsequently relative weak external rotators,³ resulting in a lower ER/IR ratio, and thus increasing the risk for chronic shoulder pain.

The role of the scapula in normal shoulder function

The scapular stabilisers also play an important role in shoulder joint movement. Normal shoulder function requires smooth integration of movement not only at the glenohumeral joint but also at the scapulothoracic, acromioclavicular (AC) and sternoclavicular (SC) joints. This integrated movement is referred to as 'scapulohumeral rhythm'. Scapulohumeral rhythm should be smooth, coordinated and symmetrical. Full upper limb elevation requires three-dimensional rotation of the scapula and the clavicle as coupled movements into upward rotation, posterior tilting and adequate internal or external rotation, depending on the plane of the moving humerus, which ensures that the coracoacromial arch is removed from the path of the greater tuberosity of the elevating humerus, thus avoiding potential impingement.

Scapular control also enhances joint stability at greater than 90° of abduction by placing the glenoid fossa under

the humeral head, where stability is assisted by the action of the deltoid muscle. The muscles controlling the scapula are the trapezius (all three portions), serratus anterior (upper and lower portions), rhomboids, levator scapulae and pectoralis minor. These muscles work in coordinated patterns called force couples to control three-dimensional scapular motion.

The main upward rotation force couple involves the upper trapezius/lower part of the serratus anterior, with a stabilising role for the middle and lower trapezius. The rhomboids and levator scapulae are supposed to lengthen in order to allow sufficient scapular upward rotation. Anterior/posterior tilt involves the lower trapezius/lower part of the serratus anterior force couple, and requires relative relaxation of the pectoralis minor. Together with the levator scapulae and the rhomboid, pectoralis minor is considered to have a rather postural function, keeping the scapula steady on the thoracic wall in rest and during movement. Interestingly, these muscles (pectoralis minor, levator scapulae and rhomboid) are very often tight rather than weak in patients with shoulder pain, and need to be addressed aiming at relaxation rather than strengthening.

The third scapular movement, internal and external rotation, is controlled by a complex force couple consisting of both parts of the serratus anterior versus all three parts of the trapezius. All three movements, and their working force couples are summarised in Figure 24.2. A stable scapula provides a base for the muscles arising from the scapula and acting on the humerus, allowing them to maintain their optimal length-tension relationship.

There is a body of evidence suggesting altered scapular rhythm or scapular dyskinesis, consisting of abnormal scapular positioning and/or movement patterns in patients suffering from shoulder pain, although the cause-consequence relationship is still under debate. Scapular dyskinesis has been found to be related to shoulder instability, impingement and stiff shoulders in a variety of patient populations, and is identified as a primary risk

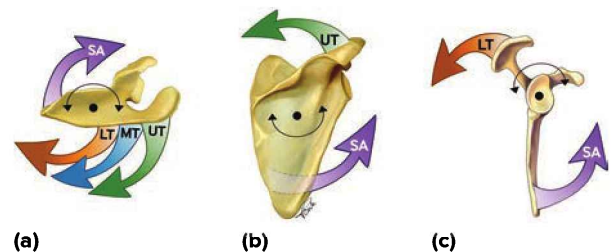


Figure 24.2 Force couples controlling scapular motions (a) Internal/external rotation (b) Upward/downward rotation (c) Anterior/posterior tilting

SA = serratus anterior, UT = upper trapezius, MT = middle trapezius, LT = lower trapezius

factor for shoulder injury in handball³ and rugby players.⁴ Scapular malpositioning and abnormal muscle patterning have been associated with experimentally induced shoulder pain, thus suggesting scapular dyskinesis might be secondary to shoulder pain.⁵ Abnormalities of scapulohumeral rhythm are most commonly due to shortening of the scapulohumeral or glenohumeral muscles, or weakness and/or poor motor control of the scapular stabilisers (with or without weakness of the rotator cuff muscles).

CAUSES OF SHOULDER PAIN—OVERVIEW

Most athletic shoulder injuries are the result of one of two mechanisms: (1) an acute external force applied on the shoulder complex (macrotrauma) or (2) repetitive overhead activity and overuse (microtrauma).

Acute versus overuse shoulder pain

The shoulder is susceptible to traumatic injuries such as dislocations, subluxations, acromioclavicular joint sprains, and soft tissue injuries, in particular in collision and contact sports. However, most injuries result from repetitive overuse mechanisms, due to overload, aberrant overhead throwing biomechanics and dysfunctional adaptations to the sport, leading to chronic symptoms like impingement and bursitis.⁶ Moreover, the shoulder is predisposed to athletic injury because of the large amount of mobility of the glenohumeral joint, allowing powerful throwing and smashing, but putting the shoulder at risk for injury due to the inherently poor glenohumeral stability.

A list of possible causes of shoulder pain is presented in Table 24.1

Impingement

Impingement is one of the most frequently described pathologic shoulder conditions in general practice and in sports medicine. Early literature described impingement as a pathology or a diagnosis but today impingement is considered to be a cluster of symptoms, rather than a pathology itself. Various investigations have confirmed the association between impingement symptoms and a variety of underlying pathologic mechanisms. Rotator cuff pathology, scapular dyskinesis, shoulder instability, biceps pathology and superior labrum anterior to posterior (SLAP) lesions, and glenohumeral internal rotation deficit (GIRD) or posterior shoulder stiffness have been suggested to cause shoulder impingement symptoms. In addition, thoracic posture and mobility are considered to be key factors in the development of shoulder pain.

The literature describes two types of impingement: subacromial and internal. Subacromial or external impingement is the mechanical encroachment of the soft tissue (bursa, rotator cuff tendons) in the subacromial space between the humeral head and the acromial arch (Fig. 24.3). This encroachment particularly takes place in the midrange of motion, often causing a 'painful arc' during active abduction.

Internal impingement comprises encroachment of the rotator cuff tendons between the humeral head and the glenoid rim. Based on the location of the impingement, anterosuperior and posterosuperior glenoid impingement have been described. Posterosuperior glenoid impingement involves mechanical encroachment of the rotator cuff tendons, particularly the supraspinatus and infraspinatus, between the greater tubercle of the humerus and the

Table 24.1 Causes of shoulder pain

Common	Less common	Not to be missed causes outside the shoulder girdle
Rotator cuff pathology (strain, tear, tendinopathy)	Other muscle tear (pectoralis major, long head of biceps (LHB))	Somatic referred pain (cervical and thoracic spine, myofascial structures)
Glenohumeral instability (traumatic, atraumatic, congenital)	Adhesive capsulitis	Tumour (bone tumours in proximal humerus)
Biceps related pathology (SLAP labral tears, biceps tendinopathy, tenosynovitis)	Neurovascular entrapment or traction injury (suprascapular nerve, long thoracic nerve, axillary nerve, axillary vein compression, thoracic outlet syndrome)	Visceral referred pain (e.g. diaphragm, gall bladder, heart, spleen, apex of lungs)
Posterior shoulder stiffness (GIRD)	Fractures (scapula, humerus, stress fracture of coracoid process)	
Scapular pathology and dysfunction, including AC and SC joint pathology (sprain, fracture, dislocation, clavicular fracture)	Snapping scapula	

SLAP = superior labrum anterior to posterior; GIRD = glenohumeral internal rotation deficit; AC = acromioclavicular; SC = sternoclavicular

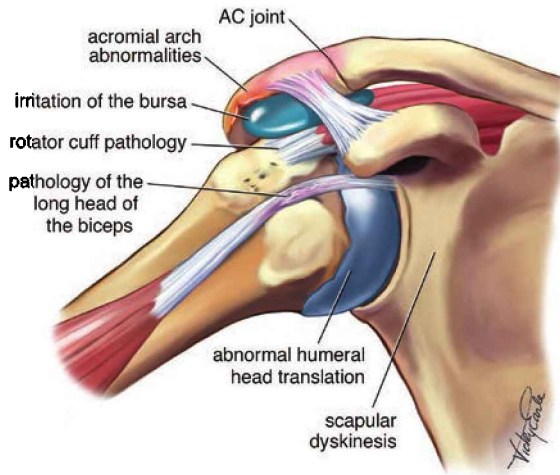
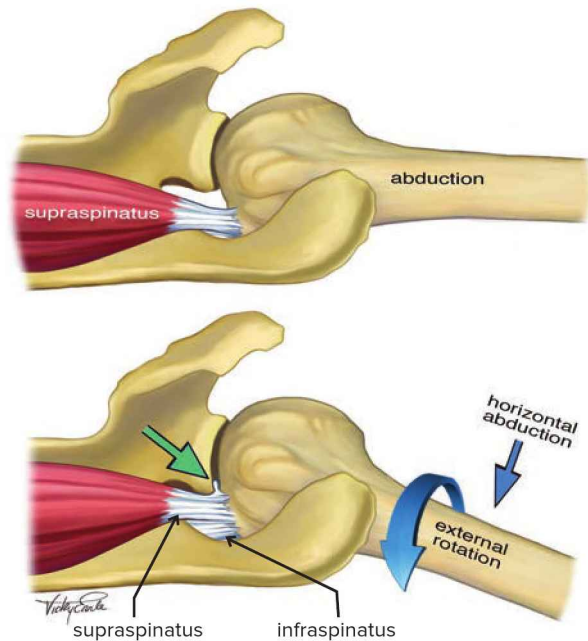


Figure 24.3 Subacromial impingement

posterosuperior rim of the glenoid (Fig. 24.4a). This occurs during the late cocking position of throwing, which is maximal external rotation, horizontal abduction, and depending on the specific sport discipline, a certain amount of elevation. Because of the specific position of this internal impingement, it is considered to be the primary cause of chronic shoulder pain in the overhead athlete. Internal impingement particularly occurs when the humeral shaft goes beyond the plane of the body of the scapula during the cocking position of throwing. Under normal circumstances, the scapula goes into retraction simultaneously with the horizontal abduction movement of the humerus. When the body of the scapula and the humeral shaft fail to remain in the same plane of movement during the cocking phase of throwing, encroachment of the rotator cuff tendons between the humeral head and the glenoid rim may cause internal impingement symptoms. This phenomenon is called 'hyperangulation' (Fig. 24.4b).⁷

Besides the classification of impingement based on the site of encroachment, impingement is also classified based on the cause of the problem, dividing it into primary versus secondary impingement. In primary impingement, a structural narrowing of the subacromial space causes pain and dysfunction, such as acromioclavicular arthropathy, type II or III acromion, subacromial bone spurs, or swelling of the soft tissue in the subacromial space, for instance the subacromial bursa. The bony subacromial abnormalities result from either a congenital abnormality (os acromiale) or osteophyte formation. In addition, the under surface of the acromion may be abnormally beaked, curved or hooked (Fig. 24.5). However, abnormalities are not necessarily associated with clinical symptoms.⁸

In secondary impingement, there are no structural obstructions causing the encroachment, but rather



(a)



(b)

Figure 24.4 Internal impingement in overhead athletes (a) Posterosuperior impingement of the supraspinatus and infraspinatus tendons (b) Hyperangulation of the humerus in the late cocking phase of throwing is a common mechanism of internal impingement

functional problems occurring only in specific positions. Rotator cuff weakness, instability and scapular dyskinesis are possible underlying functional deficits, leading to secondary impingement symptoms. Secondary impingement may occur in the subacromial space as well as internal in the glenohumeral joint.

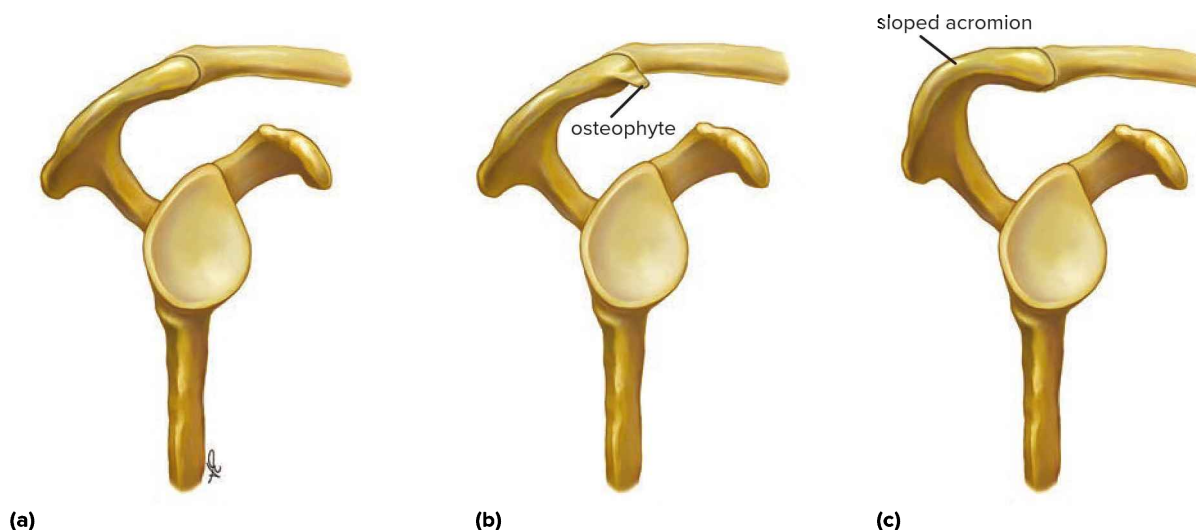


Figure 24.5 Acromial shapes. Abnormalities are not necessarily associated with clinical symptoms (a) Normal acromion (b) Acromion with anterior osteophyte (c) Congenital sloped acromion

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Primary impingement might be considered as structural impingement, whereas secondary impingement is in most cases functional impingement.

In view of the assumption that impingement symptoms may be the result of various underlying pathologies, it is important to describe the biomechanical relationship between these symptoms and shoulder diagnoses. Rotator cuff pathology may be associated with impingement symptoms in primary as well as secondary impingement. In primary impingement, swelling of the injured rotator cuff tendons causes the narrowing of the subacromial space; in secondary impingement, dysfunction of the rotator cuff (whose function is amongst others performing a caudal glide of the humeral head during elevation in order to avoid impingement) results in more cranial migration of the humeral head, thus causing secondary impingement.

Scapular dyskinesia has also been described in relation to impingement symptoms. The rationale behind this association is that during arm elevation, impingement may occur if the scapula insufficiently follows the humeral head movements because of a lack of upward rotation, posterior tilting and external rotation, thus clearing the acromial arch or the glenoid rim from impinging the soft tissue.

The association between impingement symptoms and shoulder instability is well established. Excessive humeral head translation, based on capsular laxity and instability, cause temporal narrowing of the subacromial space, thus leading to impingement symptoms and pain.

Since the biceps plays an important role in shoulder stability and function, biceps pathology may cause secondary impingement symptoms. Indeed, biceps tendon problems (tendinopathy or tenosynovitis) as well as SLAP lesions (labral lesions at the site of origin of the long head of the biceps) compromise optimal shoulder function, and may result in impingement.

Glenohumeral internal rotation deficit, often referred to as GIRD, is a sport-specific adaptation of posterior shoulder structures to chronic excessive overload of these structures during frequent throwing. There are several theories concerning the occurrence and development of GIRD. A first hypothesis is that GIRD results from contracture or shortening of the posterior capsule. Other researchers believe that GIRD begins in the early years of overhead throwing with a bony adaptation of the humerus, resulting in altered humeral torsion. A third hypothesis regarding the cause of GIRD is muscle hypertonicity in the external rotators due to frequent eccentric loading. GIRD may cause aberrant glenohumeral kinematics and thus result in impingement symptoms.

CLINICAL APPROACH

A complete clinical assessment involves:

1. history
2. observation and pre-examination palpation
3. basic clinical exam (BCE) and tissue-specific palpation
4. special tests: diagnostic testing and symptom modification tests
5. screening of the kinetic chain
6. objective measurements and outcome scores.

History

Endeavour to determine the exact site of the patient's pain; this can be difficult. Although acromioclavicular (AC) joint pain (C4 dermatome) and bicipital pain (along the bicipital groove and the line of the muscle belly) are well localised, the pain of most other shoulder pathologies is more diffuse. The onset of shoulder pain may be either acute, for example, a dislocation, subluxation or rotator cuff tear, or insidious, such as rotator cuff tendinopathy. Although most patients with chronic impingement-related shoulder pain report pain distribution in the deltoid area (and the C5 dermatome), overhead athletes with internal impingement and tissue involvement of the infraspinatus accentuate posterior shoulder pain in their history.

For acute injuries, identify the position of the shoulder at the time of injury. If the arm was wrenched backwards while in a vulnerable position, it suggests anterior dislocation or subluxation. A fall onto the point of the shoulder can cause AC joint injury. In chronic shoulder pain, the activity or position that precipitates the patient's pain should be noted, such as the cocking phase of throwing or the pull-through phase of swimming.

Note the severity of the pain, aggravating and easing factors, and the effect of the pain on activities of daily living and sporting activity. Night pain is very common in rotator cuff dysfunction and adhesive capsulitis. In order to identify the actual irritability of the injured tissue, ask (a) if the patient can sleep on the injured shoulder, (b) if the pain is only at the shoulder region, and not referring into the upper arm or elbow, and (c) if there is no pain at rest. If the answer is yes on all three questions, irritability is low, and controlled loading of the structures will be allowed. However, if the answer to one or two of the three questions is no, high irritability is present, meaning initial treatment will mainly have to focus on pain control and tissue protection.⁹

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Shoulder pain should be considered irritable if the patient cannot sleep on the painful shoulder, if their pain extends into the upper arm or elbow, or if they have resting pain. Initial treatment should then focus on pain control and tissue protection.

Sensory symptoms such as numbness or pins and needles should be noted as well as any episodes of 'dead arm'. In a baseball pitcher this suggests a labral injury. He or she may report catching and locking, or inability to develop normal speed in the action.

Inquire as to past or present problems elsewhere in the kinetic chain such as knee or ankle sprains or lower back pain. Also, clarify the exact treatment for previous local or

distant problems. Look for predisposing factors. For example, a training diary may reveal excessive load on the region.

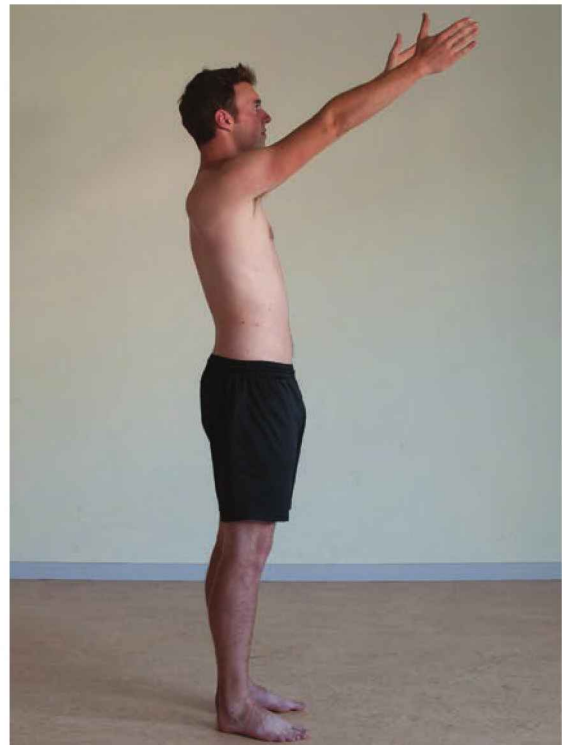
Physical examination

Examination should begin with observation of the patient from the front, from behind, and from the side. Check for asymmetry in muscle bulk, scapular position, shoulder height, spinal curves, head position, general posture, and AC and sternoclavicular (SC) joint protuberances. Next, a pre-examination palpation should be performed, checking for muscle atrophy, swelling, temperature and general trophic condition of skin. Tissue-specific palpation should be avoided prior to the examination, as this might influence the results of the clinical examination.

A basic clinical examination should then be performed, consisting of active and passive movements and resistance tests.

Active movements

These consist of forward flexion in the sagittal plane, abduction in the frontal plane, elevation in the scapular plane, functional external rotation (hand to neck) and functional internal rotation (hand behind back) (Fig. 24.6). Check for range of motion (ROM), pain and quality of



(a)

Figure 24.6 Basic clinical examination—active movements
(a) Forward flexion



(b)



(d)



(c)



(e)

Figure 24.6 (cont.) (b) Abduction (c) Elevation in the scapular plane (d) Functional external rotation—hand to neck (e) Functional internal rotation—hand behind back

movement. A specific pain pattern can be found in the painful arc, in which the patient experiences pain in the midrange of motion, and not in the extremes. Specifically observe scapular movements. Scapular dyskinesis may be observed during all active movements. Depending on the specific kind of dyskinesis, they may be divided into type I (prominent inferior angle of the scapula), type II (prominent medial border of the scapula), and type III (prominent superior medial angle of the scapula). In addition, check for a possible protracted scapular position and rounded shoulder posture.

Passive examination

This consists of forward flexion, abduction, external rotation in neutral position, internal rotation from neutral, and horizontal adduction (Fig. 24.7). Check for ROM, pain and end feel. Be aware of the specific pattern in case of adhesive capsulitis: both rotations limited, as well as abduction in the frontal plane, and a stiff end feel. Loss of ROM and/or pain during passive examination indicate articular, capsular or ligamentous injury or stiffness in the surrounding muscles.

Resistance examination

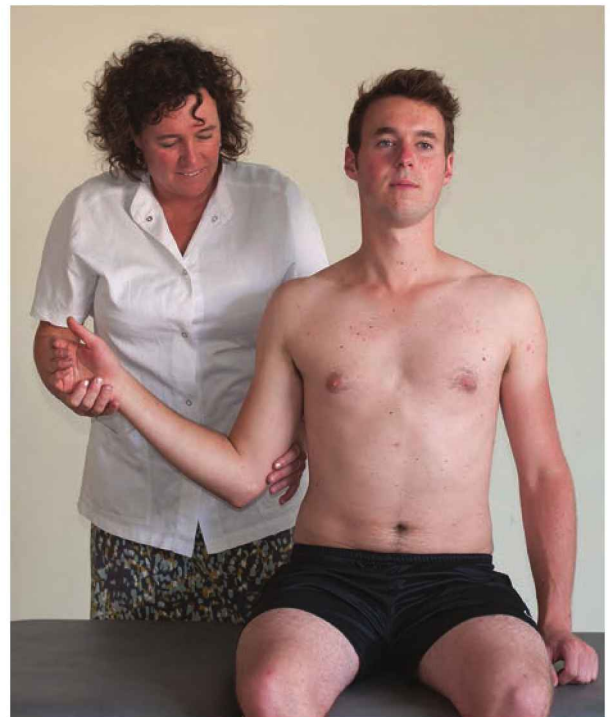
In this stage, isometric resistance is given against forward flexion, extension, abduction and adduction, external and internal rotation, elbow flexion and extension, all with



(b)



(a)



(c)

Figure 24.7 Basic clinical examination—passive movements (a) Forward flexion (b) Abduction (c) External rotation



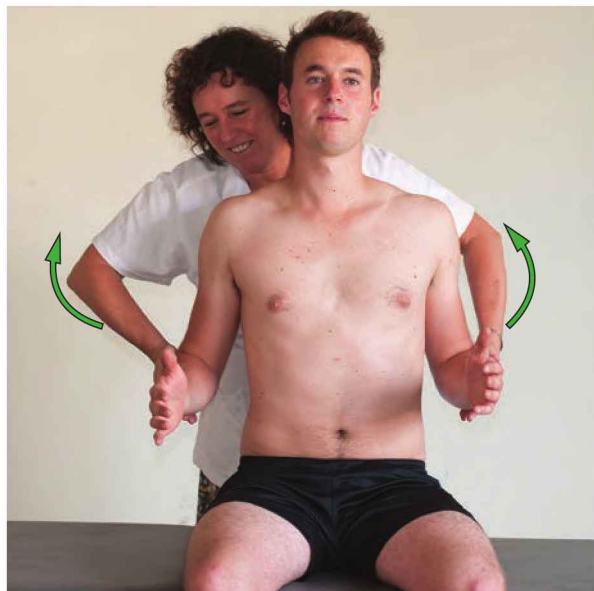
(d)



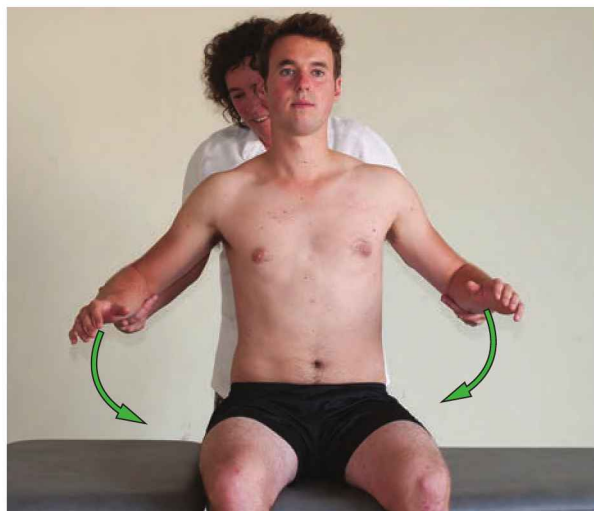
(e)

Figure 24.7 (cont.) (d) Internal rotation (e) Horizontal adduction

the shoulder in the neutral positions (Fig. 24.8). Pain and/or weakness are noticed, and might reflect injury to the musculotendinous system. Pain without weakness often is the result of tendinopathy, whereas weakness is the result of muscle tears. However, be aware that the patient might exhibit protective muscle inhibition in case of an injury with high irritability, possibly leading to a false diagnosis of muscle tear. In that case, resistance tests will be weak and painful. Weakness without pain might be the result of a complete but asymptomatic tear (e.g. external rotation



(a)

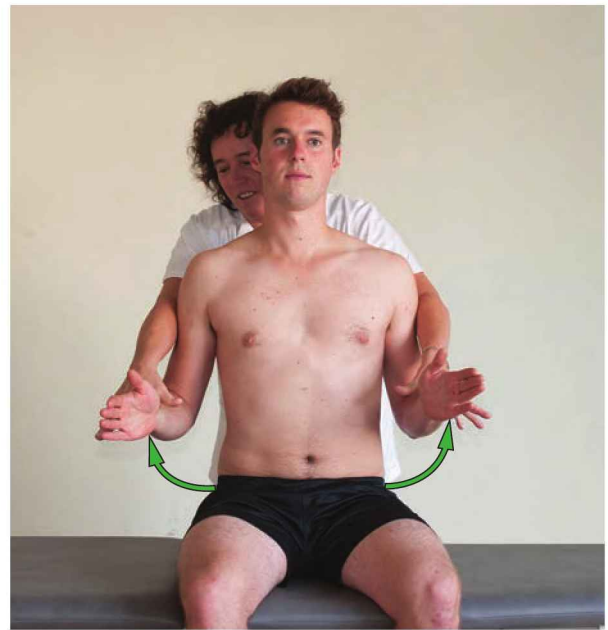


(b)

Figure 24.8 Basic clinical examination—resistance (a) Abduction (b) Adduction



(c)



(e)



(d)



(f)

Figure 24.8 (cont.) (c) External rotation (d) Internal rotation (e) Elbow flexion (f) Elbow extension

in a total tear of the infraspinatus), sport-specific weakness of the antagonist (e.g. external rotators in overhead athletes) or a neurological problem (e.g. weak abduction due to an axillary nerve injury).

After the basic clinical examination, and based on the hypothesis of tissue injury, tissue-specific palpation may be performed. In overhead athletes, check routinely for tenderness of the supraspinatus, infraspinatus, subscapularis,

AC joint line, and the LHB tendon, as well as myofascial pain in the upper trapezius, levator scapulae, pectoralis minor and rhomboids. Palpation has been shown to have similar sensitivity and specificity as to some clinical tests; however, accuracy depends upon the palpation skills of the examiner. Since the special tests, described below, often tend to irritate or impinge the tendons, it is advisable to perform palpation after the basic clinical examination and before the special tests. Repeat the tissue-specific palpation at the very end of the examination.

Special tests—diagnostic testing and symptom modification tests

Although a large number of special tests are described for examination of the shoulder, it is not feasible to undertake all of them in every examination. Here we present and discuss an algorithm for clinical reasoning and physical examination testing to assist the clinician in the assessment of impingement-related shoulder pain (Fig. 24.9).¹⁰ Clinical tests are selected based on the available evidence regarding their diagnostic accuracy

(Table 24.2); however, given the large number of available clinical tests, also based on the examiner's preference. Any test in the algorithm may be substituted by another one, as long as the purpose of the test and its diagnostic value are comparable. In general, in the interpretation of clinical tests, latest evidence and systematic reviews suggest it is favourable to interpret a 'best-test combination' instead of a solitary test.

Impingement tests

Of the various provocative impingement tests, the most popular are the Jobe, Hawkins and Neer tests. In the Jobe test or empty can test, the shoulder is put in 90° elevation in the scapular plane in maximal internal rotation (empty can position) and manual resistance is given against further elevation (Fig. 24.10a). The test is positive for subacromial impingement if the patient reports pain. The test will be negative if the patient has posterosuperior glenoid impingement.

In the Hawkins test, passive internal rotation is performed with the shoulder in 90° of forward flexion (Fig. 24.10b).

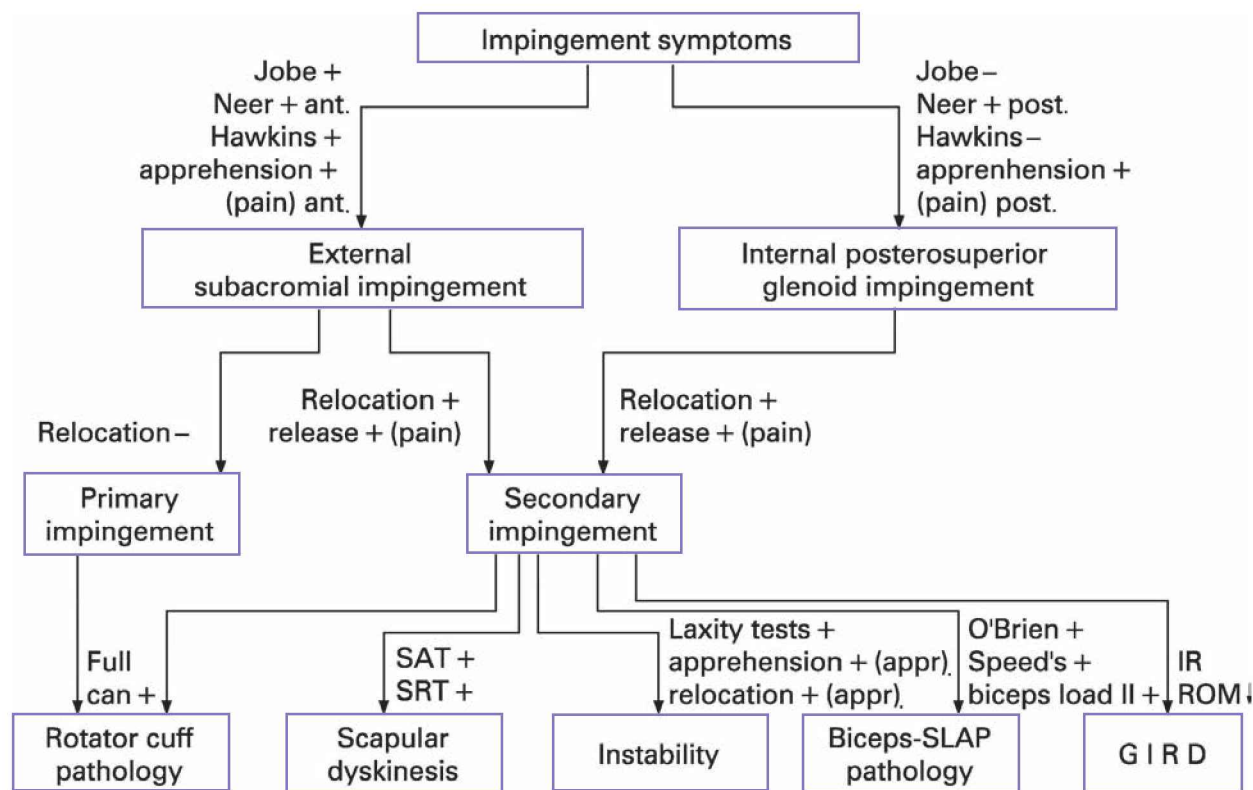


Figure 24.9 Clinical reasoning and physical examination algorithm for impingement-related shoulder pain¹⁰

+ = positive test; - = negative test; ant = anterior; post = posterior; SAT = scapular assistance test; SRT = scapular retraction test; SLAP = superior labrum anterior to posterior; GIRD = glenohumeral internal rotation deficit; IR = internal rotation; ROM = range of motion; appr = apprehension REPRODUCED WITH PERMISSION FROM *BRITISH JOURNAL OF SPORTS MEDICINE*

Table 24.2 Clinical utility of selected assessment findings for shoulder pain

Assessment finding	What is a positive finding?	Sensitivity	Specificity	LR–	LR +	Accuracy
Impingement						
Subacromial: Jobe + Hawkins + Neer + Painful arc + resistance ER	Pain, cut off 3/5 +	75	74	0.34	2.93	79
Internal: posterior impingement sign ('apprehension position')	Posterior pain	75.5	85	0.39	3.34	78
Rotator cuff pathology						
Full can test	Weakness and pain: tendinopathy/ tear supraspinatus	59–89	54–82	0.2–0.5	1.6–3.2	65–74
Lift-off test (Gerber) + resistance IR	Weakness: tear subscapularis	50	88	0.56	4.1	97
ER lag sign	Weakness: tear infraspinatus	36	95	0.67	7.2	74
Scapular dyskinesis						
Scapular assistance test (SAT)	Decreased pain	21–24	71–72	1.07–1.10	0.75–0.83	
Scapular resistance test (SRT)	Increased strength or decreased pain	26–100	33–70	0–1.06	0.87–1.49	
Instability						
Apprehension	Apprehensive feeling	98.3	71.6	0.02	3.46	81.7
Relocation	Relief of apprehension	96.7	78	0.04	4.39	85.2
Release	Sudden (surprise) apprehension	91.7	83.5	0.10	5.55	86.4
Load and shift	↑ translation humeral head anteriorly	71.7	89.9	0.32	7.10	83.4
Sulcus	↑ translation humeral head inferiorly	28–72	86–97			
Posterior subluxation test	↑ translation humeral head posteriorly	50–91	85–100			
Slap and biceps related pathology						
Apprehension AND Speed AND O'Brien	Pain	25	92	0.82	3.13	
Speed	Pain: LHB pathology	54	81	0.58	2.77	72
O'Brien	Pain: SLAP	61	84	0.47	3.83	72
Biceps load test	Pain: SLAP	29	78	0.9	1.4	55.5
Passive compression test	Pain: SLAP	82	86	0.21	5.1	83.6

+ = positive test; ER = external rotation; IR = internal rotation; SAT = scapular assistance test; SRT = scapular resistance test; SLAP = superior labrum anterior to posterior; LR– = negative likelihood ratio; LR+ = positive likelihood ratio; LHB = long head of biceps

Table based on Hegedus et al.¹¹ and Hanchard et al.¹²

A painful test is an indication for subacromial impingement; the test will be negative in case of internal impingement.

The Neer test consists of forced maximal forward flexion in internal rotation with the scapula fixed into depression (Fig. 24.10c). Pain at the front of the shoulder is an indication for subacromial impingement, whereas patients with internal impingement will exhibit pain

at the posterior aspect of the shoulder (slightly change testing position into neutral rotational position/external rotation and fixing the scapula to front).

It is suggested that interpretation for subacromial impingement should be based on five tests: the three impingement tests described above, painful and/or weak external rotation against resistance, and a painful arc

during active elevation. Sensitivity and specificity are the highest when a cut-off value of three positive tests on five is used (Table 24.2).

Besides these impingement tests, instability tests are often used as provocation tests for impingement. In the apprehension position, the shoulder is placed passively in maximal external rotation and horizontal abduction (Fig. 24.10d). Pain at the anterior aspect of the shoulder suggests subacromial impingement; pain at the posterior aspect implies posterosuperior glenoid impingement. It might be necessary to change the position into more or less elevation to elicit subtle symptoms in the overhead athlete. In the relocation test, the investigator manually performs a dorsal glide on the humeral head in the apprehension position (Fig. 24.10e). The test is positive if the pain exhibited during the apprehension test disappears. The relocation test allows us to identify primary versus secondary impingement. If the test is positive, this means that the impingement pain is secondary, based on excessive anterior translation of the humeral head. A negative test suggests primary impingement, not dependent on the position of the humeral head.

Rotator cuff tests

To define the involvement of rotator cuff pathology in the impingement symptoms, the full can test is a valuable tool. The test consists of resisted elevation in the scapular plane with the thumbs up (Fig. 24.10f). Research has indicated that rotator cuff muscles are also highly active in this position, like in the empty can (or Jobe) test. If both tests are painful, rotator cuff pathology is present. If only the empty can test is painful, and the full can test is negative, the patient probably suffers from impingement symptoms, but not primarily related to rotator cuff pathology.

In the presence of rotator cuff pathology, the examiner can perform a number of specific tests for the supraspinatus, subscapularis and infraspinatus to find out whether one of more tendons are ruptured. Although recent research indicates isolation of rotator cuff muscles in a clinical setting is very difficult,¹³ the Gerber lift-off test, and the belly press test may be useful for subscapular tears, and the lag sign is often used to detect infraspinatus tears. In the Gerber lift-off test, the patient is asked to lift their hand from their back in a maximally internally rotated position (Fig. 24.10g); during the belly press test, the quality of movement is observed when the patient pushes his hand into his stomach (Fig. 24.10h). A positive external rotation lag sign occurs when the patient is unable to maintain the position of external rotation in a slightly abducted position and the shoulder rotates internally (Fig. 24.10i).

Scapular involvement tests

Scapular involvement in impingement-related shoulder pain may be examined by the scapular assistance test (SAT) and the scapular retraction test (SRT). The SAT, in which scapular movement quality is examined, consists of manual assistance of correct scapular movement during elevation of the arm (Fig. 24.10j). Reduction of pain during this movement compared to non-assistance confirms scapular involvement in the shoulder complaints.

In the SRT, in which scapular stability is examined, the empty can test is performed while the examiner stabilises the patient's scapula and shoulder in a position of retraction by placing the forearm along the medial border of the scapula (Fig. 24.10k). The test is positive for scapular involvement when the initial pain, present in the empty can position disappears during the SRT. Since the diagnostic value of these tests for shoulder pathology is low, they should be considered as symptom modification tests (see below) rather than diagnostic tests.

Instability tests

The clinical tests to examine shoulder instability can be divided into provocative tests and laxity tests. Commonly used provocative tests for instability are the apprehension and relocation tests, described earlier (Fig. 24.10d, e). In case of instability, patients will exhibit symptoms, such as apprehensive muscle tension and subluxation, rather than pain.

Distinct from the provocative tests, the laxity tests assess humeral translation with respect to the glenoid fossa. For anterior laxity, the load and shift test may be used (Fig. 24.10l). In this test, the humeral head is loaded in such a way as to centre it congruently within the glenoid fossa. Subsequently, the humeral head is manually shifted anteriorly, relative to the glenoid fossa, and the amount of translation is graded from 1 (translation up to, but not beyond the glenoid rim), over grade 2 (translation beyond the glenoid rim with spontaneous reduction after releasing the humeral head) to grade 3 (subluxation without spontaneous reduction). The sulcus sign allows examination of inferior laxity. In this test, inferior subluxation is produced by downward traction on the arm (Fig. 24.10m).

For posterior laxity, the posterior subluxation test is described. This test is performed by placing the patient's arm in adduction, internal rotation and 70–90° flexion. The examiner applies a posteriorly directed force along the arm, and then slowly moves the shoulder to horizontal abduction and external rotation (Fig. 24.10n). The test is considered to be positive if a clunk is felt during the latter movement, indicating the humeral head is relocated in the glenoid after being subluxed posteriorly.



(a)



(b)



(c)



(d)



(e)



(f)

Figure 24.10 Special tests for the shoulder (a) Jobe test (b) Hawkins test (c) Neer test (d) Apprehension test (e) Relocation test (f) Full can test



(g)



(h)



(i)



(j)

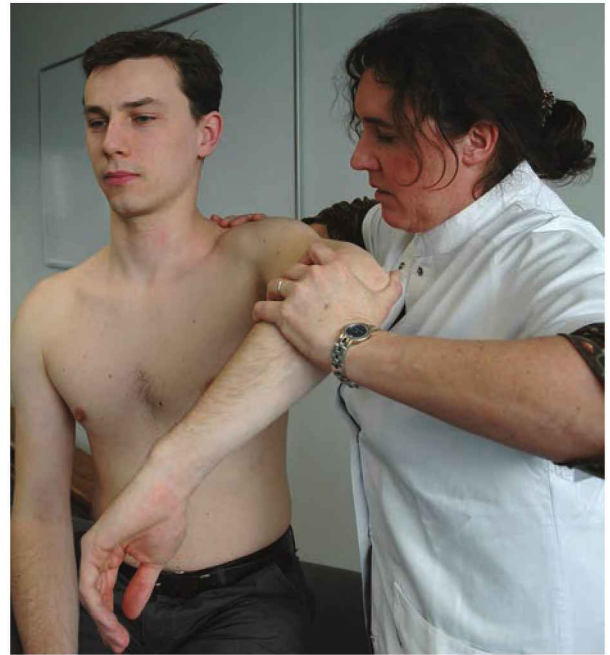


(k)

Figure 24.10 (cont.) (g) Gerber lift-off test (h) Belly press test (i) External rotation lag sign (j) Scapular assistance test (k) Scapular retraction test



(l)



(n)



(m)

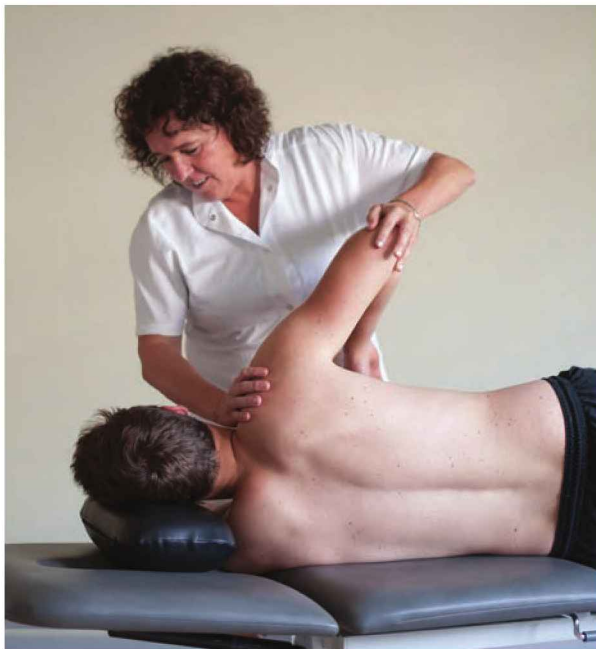


(o-i)



(o-ii)

Figure 24.10 (cont.) (l) Load and shift test (m) Sulcus sign (n) Posterior subluxation test (o) O'Brien test (i) thumb pointing down, (ii) thumb pointing up



(p)



(q)

Figure 24.10 (cont.) (p) Passive compression test (q) Biceps load II test

Biceps pathology and SLAP lesion tests

Numerous SLAP and biceps tests exist in the literature. However, we prefer the Speed test, the O'Brien test, the passive compression, and the biceps load II tests. The Speed test is performed by resisting downwardly applied pressure to the arm when the shoulder is positioned in 90° of forward flexion with the elbow extended, and forearm supinated; a positive test result should produce pain into the biceps region.

In the O'Brien test, resistance is tested with the arm forward flexed to 90° and adducted 10° with the thumb pointing down (Fig. 24.10o-i). Pain is lessened or disappears when the test is repeated in the same shoulder position but with the thumb pointing up (Fig. 24.10o-ii).

During the passive compression test, the clinician rotates the patient's arm externally with 30° of abduction and then pushes the arm proximally while extending the shoulder, which results in the passive compression of the superior labrum onto the glenoid (Fig. 24.10p).

The biceps load II test consists of applying resistance against elbow flexion with the shoulder in 120° of abduction and the elbow in 90° flexion (Fig. 24.10q). The test result is considered positive if the patient complains of pain during the resisted elbow flexion.

A best-test combination consisting of three tests (two sensitive and one more specific test) has been recommended for the interpretation (Table 24.2).¹¹

Clinical evaluation of GIRD

The assessment of glenohumeral internal rotation deficit (GIRD) is performed by measuring glenohumeral internal rotation range of motion, preferably in the supine position with the shoulder abducted 90° and the scapula stabilised against the table (Fig. 24.11). Goniometric/inclinometric



Figure 24.11 Clinical assessment of GIRD

assessment as well as interpretation of the 'end feel' is described as criteria for GIRD evaluation. A side difference of 20° is considered to be positive for GIRD. However, in view of a possible shift in rotational range of motion due to bony adaptations, GIRD may be associated with a gain in external rotation ROM. Therefore it is imperative to measure internal as well as external rotation, interpreting the total range of motion (TROM). Besides the measurement in an abducted position, additional evaluation of GIRD with the shoulder in 90° forward flexion may even give the clinician a better impression of the posterior stiffness, since there is more stretch on the posterior aspect of the shoulder.

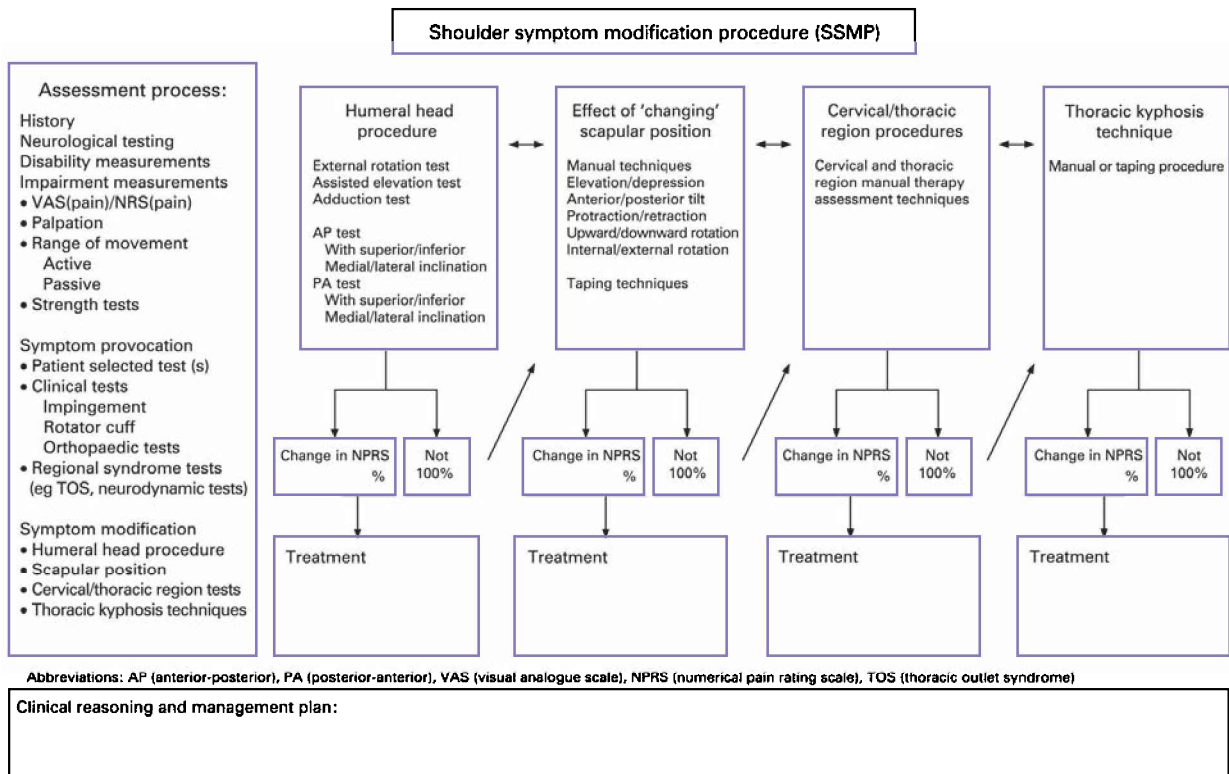
Shoulder symptom modification procedure

In view of the limited diagnostic accuracy of the existing clinical tests (Table 24.2), as well as the relevance of the rest of the kinetic chain, it is important to perform symptom modification tests. Once the movement or

activity that reproduces the shoulder symptoms has been agreed upon (and assessed during the clinical examination), the shoulder symptom modification procedure (SSMP) is applied.

The SSMP is a series of mechanical techniques that are applied while the patient performs the activity or movement that most closely reproduces the symptoms experienced by the patient (Fig. 24.12). These techniques may be applied on different levels of the kinetic chain, starting with the shoulder locally, going to the scapula (see scapular tests), thoracic and cervical spine and posture, lumbar spine and the lower extremities. Modifications may include correcting posture (for instance sitting versus standing or correcting thoracic kyphosis), ensuring correct muscle contraction (for instance scapular correction, core stability) or mechanically correct kinematics (for instance posterior glide humeral head).

The basic principles, however, are always similar: (a) define a position or movement that causes the



Patients' name, date of birth and identification

Clinician's name, signature and date

Figure 24.12 Shoulder symptom modification procedure (SSMP)

FROM LEWIS 2009.¹⁶ REPRODUCED WITH PERMISSION FROM *BRITISH JOURNAL OF SPORTS MEDICINE*

symptoms, based on the clinical exam, (b) correct the possible cause of the symptoms, based on biomechanics or arthrokinematics, and (c) observe possible change in symptoms. Based on this procedure the clinician will know the key points in the rehabilitation program and advise the patient. As stated by the authors, the SSMP is not meant as a replacement for current clinical practice but as an adjunct to help support the clinical decision-making process by identifying factors that alleviate the patient's shoulder symptoms.¹⁴⁻¹⁶

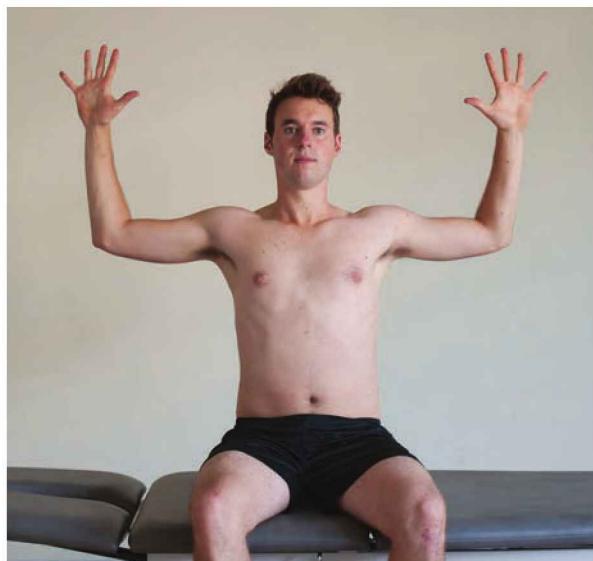
Additional shoulder tests

Depending on the history and the results of the basic clinical examination, the clinician may need to perform additional special tests to those described in the previous section. In particular, neurologic, neurodynamic and/or vascular examination might be necessary in case of a suspicion of for instance thoracic outlet syndrome (TOS) or neurodynamic upper limb pathology. The Roos test (Fig. 24.13a) is a very sensitive and specific test for TOS and easy to perform: ask the patient to open and close their hands for at least 60 seconds with the shoulders abducted and externally rotated and the shoulder girdle in depression. Motor deficits, change of colour of the hands, numbness, and slowing down of the movement are considered to be signs of TOS. For the neurodynamic examination, the basic upper limb neurodynamic tests (ULNT) may be performed, evaluating the mechanosensitivity of the nervous system in the upper quarter (Fig. 24.13b). The tests are considered positive if they provoke or reproduce the symptoms, or are in a relevant distribution to the nerve being tested. The second feature of a positive test is that the response received (sensory symptoms, ROM or resistance perceived by the examiner) is different from the asymptomatic limb, and different from what is known to be a normal response. Cervicogenic shoulder involvement needs to be assessed by a cervical spine screening and differential diagnostic tests such as the Spurling test (Chapter 23).

Screening of the kinetic chain

Regardless of the SSMP, screening of the kinetic chain is imperative in overhead athletes. Quick screening of kinetic chain variables consists of:

- observation of spinal curves and general posture (forward head posture, hyperlordosis cervical spine, hyperkyphosis and protracted shoulders, sway back, hyperlordosis or hypolordosis, flexed position of the hips, knee position into valgus or varus, foot position)
- single-leg/double-leg squat observe hip and core stability



(a)



(b)

Figure 24.13 Additional tests for shoulder pain (a) Roos test for thoracic outlet syndrome (b) Upper limb neurodynamic tests

- bilateral deep squat: observe quality of movement, repetitive movements to evaluate lower limb strength and endurance
- trunk ROM into rotation, flexion and extension
- ROM of the hips, in particular into extension and rotation.

The clinician may use some specific testing protocols, such as the functional movement screen or the 9+¹⁷ (Chapter 46) or a similar standardised screening program, looking at ROM as well as strength and stability, and at the trunk as well as the extremities.

Key outcome measures

There are numerous patient-reported outcome measures to assess functional status, treatment effect and return-to-play criteria for people with shoulder pain. The clinician's selection should be based on the specific clinical outcome to be measured (body structures, function, activity, participation), the specific pathology (instability, rotator cuff pathology) and whether disease-specific,

region-specific or generic outcome is required. Table 24.3 summarises the most commonly used outcome scales for shoulder pain patients.

INVESTIGATIONS

We discuss the principles of shoulder investigations (including diagnostic arthroscopy) as it is a critical element of efficient management of the patient with shoulder pain.

Objective measurements of the shoulder

The assessment of the ROM into rotation of the shoulder can be measured with a goniometer or an inclinometer, and in many positions of the body and the shoulder. A comprehensive reliability study showed good to excellent inter- and intra-tester reliability for a variety of test positions and equipment.¹⁸ Based on the results of this study, no specific procedure can be acknowledged as superior to another. However, the clinician has to take into account that large variability exists in literature regarding shoulder position (for instance scapular or frontal plane) and the specific method of scapular stabilisation (none, hand on shoulder top, or specific fixation of coracoid), and therefore be consequent in repeated measurements. Based on the above-mentioned reliability study and in view of optimal standardisation of body and shoulder position, the authors advise the procedure supine, arm 90° in frontal plane, palpation of the coracoid process, performed with digital inclinometer (Fig. 24.11).

Regarding rotator cuff strength, isokinetic (Fig. 24.14a) as well as isometric (Fig. 24.14b) and eccentric (Fig. 24.14c) strength measurements may be performed. Absolute side differences as well as muscle balance ratio between external and internal rotators are to be interpreted. In general, with respect to cut-off values distinguishing a healthy shoulder from a shoulder at risk, an isokinetic ER/IR ratio of 66% or an isometric ER/IR ratio of 75% (when measured in neutral or in abduction 90° and external rotation 90°), or $\pm 100\%$ (when measured in 90° abduction and 0° rotation) and 60–80% (when measured in 90° abduction and 90° external rotation) is advised, with a general rotator cuff strength increase of 10% of the dominant throwing side¹⁹ compared to the non-dominant side. The preferential functional eccentric/concentric ratio (eccER/concIR) depends on the speed and specific procedure tested. For the eccentric test with the hand-held dynamometer (HHD), values slightly higher than 100% (114–121%) were found in a sample of 201 adult overhead athletes from volleyball, tennis and handball.¹⁹ All procedures have been found to show high to excellent reliability.

Scapular performance or behaviour can be evaluated by measuring scapular inclination (upward rotation), scapular muscle strength (isokinetic or isometric testing) and muscle length (pectoralis minor).

Scapular inclination may be measured using an inclinometer in several arm elevation positions (Fig. 24.15). In



(a)

Figure 24.14 Rotator cuff strength testing (a) Isokinetic testing

continued



(b)



(c)

Figure 24.14 (cont.) (b) Isometric testing with a hand-held dynamometer (c) Eccentric testing with a hand-held dynamometer—the assessor performs an internal rotation of the shoulder while the patient maximally resists against this pressure



Figure 24.15 Measurement of scapular inclination using an inclinometer along the spine of the scapula

general, an upward inclination of approximately 60° should be reached at full active arm elevation.

For the scapular muscles proper intermuscular and intramuscular balance should be assessed. Isokinetic protraction/retraction ratio (Fig. 24.16) should be 100% in a healthy population, with slight changes in overhead athletes who have slightly stronger protractors. When measured isometrically with an HHD, it is advisable to use standardised positions²⁰ and to be consistent doing a make or a break test, since results might differ significantly. In bilateral sports (swimming, rowing, gymnastics) there should be no side



(a)

Figure 24.16 Isometric strength testing of the scapular muscles (a) Protraction



(b)

Figure 24.16 (cont.) (b) Retraction

differences in scapular muscle strength, in one-handed overhead sports an increase of scapular muscle strength of 10% is advised on the dominant side. In particular, the lower trapezius and serratus anterior should receive special attention, since these muscles are shown to be susceptible to weakness in injured athletes. However, a statement saying that scapular behaviour should be symmetrical in overhead athletes is not supported by data. On the contrary, in volleyball, as well as in handball players, asymmetry in resting scapular posture is often found. Therefore clinicians should be aware that some degree of asymmetry in resting scapular posture is often found. It should not be considered automatically as a pathological sign but rather an adaptation to sports practice and extensive use of the upper limb.

Tightness of the pectoralis minor may be clinically observed by evaluating shoulder position with the patient supine (with more protracted shoulder position in case of short pectoralis minor) and objectively measured by determining the distance between the coracoid process and the fourth rib at the sternum (Fig. 24.17a). When normalised to body height (pectoralis minor index (PMI) = pectoralis length/body height), the PMI should not be below 7.65, indicating a short pectoralis minor.²¹

Functional performance tests for shoulder function are gaining interest, although not yet fully explored in clinical practice. The seated medicine ball throw (SMBT) (Fig. 24.17b) and the Y-balance test for the upper extremity (Fig. 24.17c) have been described for evaluation of the front chest throwing performance and static weight-bearing shoulder stability, respectively. However, normative data and cut-off values for injury prevention and return to play are lacking. Recently, it was shown that the SMBT and isokinetic shoulder, and elbow strength correlate highly, whereas no significant correlation between the Y-balance test and these isokinetic strength measures was found.²²



(a)



(b)



(c)

Figure 24.17 (a) Measurement of pectoralis minor length (b) Seated medicine ball throw (c) Y-balance test for the upper extremity

Table 24.3 Patient-reported outcome measures for shoulder pain**Generic shoulder pain****Disabilities of the Arm, Shoulder and Hand questionnaire (DASH)**

- Self-administered questionnaire evaluating symptoms and function of the entire upper extremity
- 30 items
- Two optional additional modules available for work and sports/performing arts
- Score range 0–100
- Test-retest reliability: 0.93–0.98^{23–27}
- Construct validity: shoulder Pain and Disability Index (SPADI) 0.55–0.93^{24, 28, 29}
- MDC95%: 7.9–14.8^{24, 26, 30}
- MCID: 10.2²⁶
- Available online in a range of languages: www.dash.iwh.on.ca

QuickDASH

- Self-administered questionnaire evaluating symptoms and function of the entire upper extremity²³
- 11 items
- Test-retest reliability: 0.90–0.94^{31–33}
- Construct validity: SPADI: 0.84³⁴
- MDC95%: 13.3³³
- MCID: 8³³
- Available online in a range of languages: www.dash.iwh.on.ca

Simple Shoulder Test (SST)

- Self-administered questionnaire that evaluates functional disability of the shoulder
- 12 items³⁵
- Score range 0–100
- Test-retest reliability 0.97³⁶–0.99³⁷
- Construct validity: SPADI 0.74³⁵–0.80³⁸/DASH: 0.72³⁹
- MDC95%: 32.3³⁸
- MCID: 17.1–25⁴⁰
- Available online in a range of languages: www.orthop.washington.edu

Shoulder Pain and Disability Index (SPADI)

- Self-administered questionnaire evaluating symptoms and function of the shoulder
- 13 items
- Two optional additional modules available for work and sports/performing arts
- Test-retest reliability 0.84–0.95^{26, 30, 36, 41, 42}
- Construct validity: DASH 0.93,²⁸ 0.55²⁹ and 0.88⁴¹
- MDC95%: 13.2–21.5^{26, 38, 41, 42}
- MCID: 13.2–23.1^{26, 38, 43}
- Available online: www.workcover.com

Rotator cuff pathology**Western Ontario Rotator Cuff Index (WORC)**

- Self-administered, condition-specific questionnaire evaluating quality of life
- 21 items
- Score range 0–2100
- Test-retest reliability 0.84–0.96^{44–46}
- Construct validity: DASH 0.49⁴⁵ and 0.63⁴⁶
- MDC95%: 19.1⁴⁵
- MCID: 245.26⁴⁷
- Available online: www.secec.org/pages/r-d/assessments-scores.php

Shoulder instability**Western Ontario Shoulder Instability Index (WOSI)**

- Self-administered questionnaire evaluating quality of life of patients with symptomatic shoulder instability
- 21 items
- Score range 0–2100
- Test-retest reliability: 0.87–0.98^{48–51}
- Construct validity: 0.77⁴⁸
- MDC95%: 483⁵²
- MCID: NA
- Available online: www.secec.org/pages/r-d/assessments-scores.php

Shoulder instability**Kerlan–Jobe Orthopaedic Clinic Shoulder and Elbow Score**

- Self-administered 10 item questionnaire that evaluates throwing specific symptoms and function in throwing athletes
- 10 items (five on symptoms and five on shoulder function)
- Test-retest reliability: 0.88⁵³
- Construct validity: NA
- MCD95% NA
- MCID: NA

NA = not available

Radiography

Plain X-rays are important in the diagnosis of shoulder abnormalities. Routine views (AP with internal and external rotation and axillary lateral) provide a good overview of the region. A 'true' AP view is useful for assessing joint space narrowing, (i.e. arthritis). In cases of trauma, an adequate axillary view may not be possible and a true lateral film must be obtained to exclude dislocation. The following can be identified on plain films: calcific tendinopathy, glenohumeral joint arthritis, impingement (sclerosis of anterior and/or lateral acromion, sclerosis of greater tuberosity), proximal humeral head migration (severe rotator cuff dysfunction) and fractures.

Special views have been described to evaluate instability and impingement. Supraspinatus outlet views and down-tilted acromial films are obtained to evaluate impingement. In cases of instability, special views such as the West Point view or the Stryker notch view are used to better detect Bankart and Hill-Sachs' lesions.

Arthrography

Detailed anatomical information is obtained when arthrography (joint injected with dye) is combined with CT of the shoulder (CT arthrogram) or MR arthrography (see below). This examination gives excellent detail of capsular attachments and of the labrum. Small avulsion fractures of the glenoid rim (Bankart lesion) and the humeral head (Hill-Sachs' lesion) are clearly defined.

Ultrasound

High-resolution ultrasound, in the hands of an experienced operator, is a reliable non-invasive technique for imaging the rotator cuff and adjacent muscles, the bursae and the long head of the biceps muscle. The examination may be performed as a static or dynamic investigation. Tendon swelling, thickening of the bursa, abnormal fluid collection or calcific tendonitis may be detected, as may a partial or complete rotator cuff tear. It is important to define the size and location of the tear, and if there is any supraspinatus muscle atrophy (indicating a chronic and usually irreparable rotator cuff tear). If there is a partial thickness tear it is important to determine the thickness of the tear—as a percentage of the thickness of the tendon—as tears over 50% usually progress and are less likely to respond to non-operative treatment. It is also important to determine if the partial thickness tear is on the bursal side, under surface or intrasubstance. A dynamic examination performed while the patient is actively abducting the shoulder may confirm the presence of impingement (see tendinopathy management).

Magnetic resonance imaging

MRI allows multiplanar, non-invasive examination of the shoulder and is used to detect a rotator cuff tear. Bone detail is not defined as well as with CT and examination with shoulder movement is not possible. MR arthrogram with contrast is well suited to evaluate labral tears or instability.

Diagnostic arthroscopy

Arthroscopy of the shoulder, as well as being therapeutic, can provide useful diagnostic information. Shoulder arthroscopy permits inspection of the glenohumeral joint and the subacromial space in turn. Arthroscopy of the glenohumeral joint cavity is particularly useful as it:

- enables inspection of the glenoid labrum for evidence of a Bankart lesion or a SLAP lesion
- permits assessment of the state of the articular cartilage
- will demonstrate the presence of a Hill-Sachs' lesion
- allows inspection of the shoulder capsule and synovium (a red synovium and thickened capsule are characteristic of adhesive capsulitis)
- will identify a drive-through sign for laxity
- permits inspection of the under surface of the rotator cuff tendons, the biceps tendon and the subacromial bursa
- enables inspection and probing of the bursal surface of the rotator cuff.

Arthroscopy of the subacromial space allows assessment of:

- bursitis
- coracoacromial ligament ossification (spur formation)
- lateral spurs
- os acromiale
- bursal side rotator cuff tears
- full thickness rotator cuff tears.

The examination under anaesthesia (EUA) performed in conjunction with arthroscopy may sometimes be helpful to assess the presence, direction and severity of shoulder laxity, and to assess shoulder range of motion.

It is important to remember that these sophisticated investigations are only an adjunct to the clinical findings. In many cases of shoulder pain, the clinical findings provide sufficient information to diagnose the cause of the shoulder pain.

GENERAL TREATMENT AND REHABILITATION GUIDELINES FOR THE MOST COMMON SHOULDER INJURIES IN ATHLETES

It is important to note that shoulder pain may have several underlying causing factors, structural and functional. The treatment approach must be based on the results of the clinical examination and the diagnostic imaging, and may comprise several of the treatment strategies described below. In case several factors are identified within the kinetic chain, it is imperative to start the treatment dealing with correcting the most proximal parts in the kinetic chain.

In general, four phases can be recognised in the rehabilitation of the overhead athlete:⁵⁴ (1) acute phase, (2) intermediate phase, (3) advanced strengthening phase, and (4) return-to-play phase. In the acute phase, the goals are to

diminish pain and inflammation, normalise motion, delay muscle atrophy and restore dynamic stability. Strengthening exercises focus on rotator cuff and scapular retractors. Functional loading is limited until full range of motion is restored. During the intermediate phase strengthening exercises progress into isotonic training of the shoulder girdle and the core, and flexibility is controlled by intensive stretching exercises, in particular of the posterior shoulder structures. The advanced strengthening phase consists of more aggressive strength training, including power and endurance enhancement. A plyometric program, endurance drills and controlled throwing are initiated. In the return-to-play phase, the athlete progressively increases the throwing program, continues flexibility drills and prepares to return to competitive throwing.

ROTATOR CUFF INJURIES

Rotator cuff injuries consist of tendinopathy and tears (complete or partial), often associated with an underlying tendinopathy.

Rotator cuff tendinopathy

The aetiology of rotator cuff tendinopathy (Fig. 24.18a) is multifactorial, and has been attributed to both extrinsic and intrinsic mechanisms. Extrinsic factors that encroach upon the subacromial space (subacromial impingement) or against the posterosuperior rim of the glenoid (internal

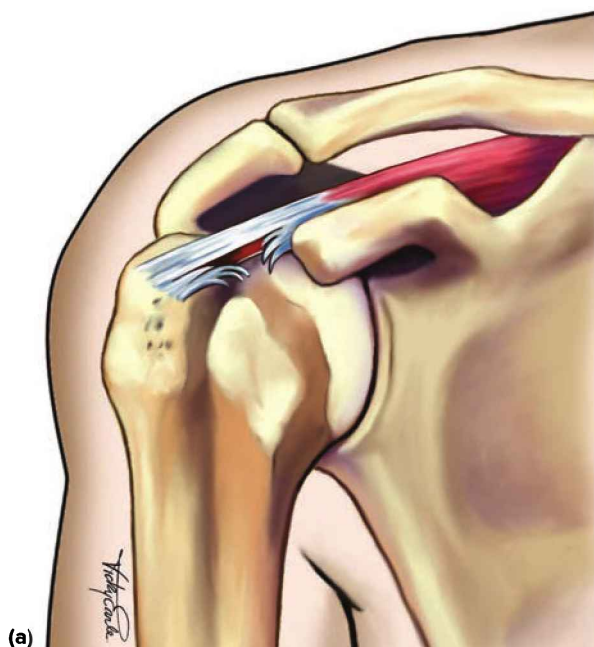


Figure 24.18 Rotator cuff tendinopathy (a) Pathology generally begins on the inferior (articular) surface of the tendon and may be associated with a partial tear

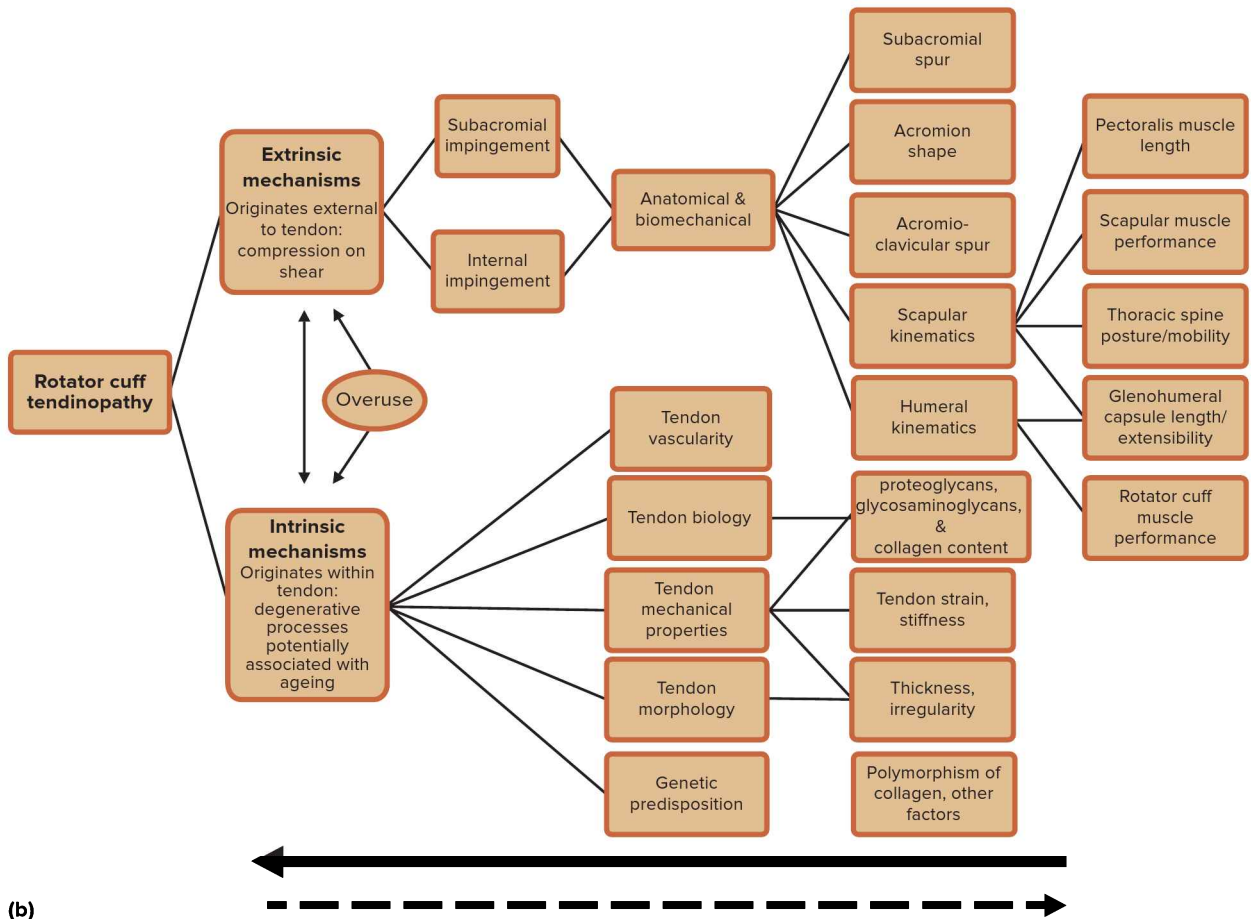


Figure 24.18 (cont.) (b) Intrinsic and extrinsic factors contributing to rotator cuff tendinopathy⁶
PG = proteoglycans; GAG = glycosaminoglycans

impingement), include anatomical variants of the acromion, alterations in scapular or humeral kinematics, postural abnormalities, rotator cuff and scapular muscle performance deficits, and decreased extensibility of the pectoralis minor or posterior shoulder. Intrinsic factors that contribute to rotator cuff tendon degradation with tensile/shear overload include alterations in biology, mechanical properties, morphology and vascularity. An overview of extrinsic and intrinsic mechanisms contributing to rotator cuff tendinopathy is given in Figure 24.18b.

Clinical features

The athlete with rotator cuff tendinopathy complains of pain with overhead activity such as throwing, swimming and overhead shots in racquet sports. Activities undertaken at less than 90° of abduction are usually pain-free. On examination, there may be tenderness over the supraspinatus tendon proximal to or at its insertion into the greater tuberosity of the humerus. If the infraspinatus is involved, there is also tenderness on the insertion dorsally

on the greater tuberosity. Active movement may reveal a painful arc on abduction between approximately 70° and 120°. Symptoms can be reproduced with impingement tests and in the apprehension position, and also the full can test and external rotation against resistance may be painful. The investigation of choice in rotator cuff tendinopathy is MRI arthrogram (Fig. 24.19). This may also identify a partial tear of the rotator cuff. Ultrasound can rule out a full thickness tear; define a partial thickness tear; identify a thickened subacromial bursa and rule in or out 'impingement' of the bursa under the lateral acromion as the arm is abducted.

Treatment of rotator cuff tendinopathy

Treatment should be considered in two parts. The first part is to treat the tendinopathy symptomatically. The patient should avoid the aggravating activity and apply ice locally. There is no level 2 evidence (Chapter 2) to support NSAIDs, ultrasound, interferential stimulation, laser, magnetic field therapy or local massage.

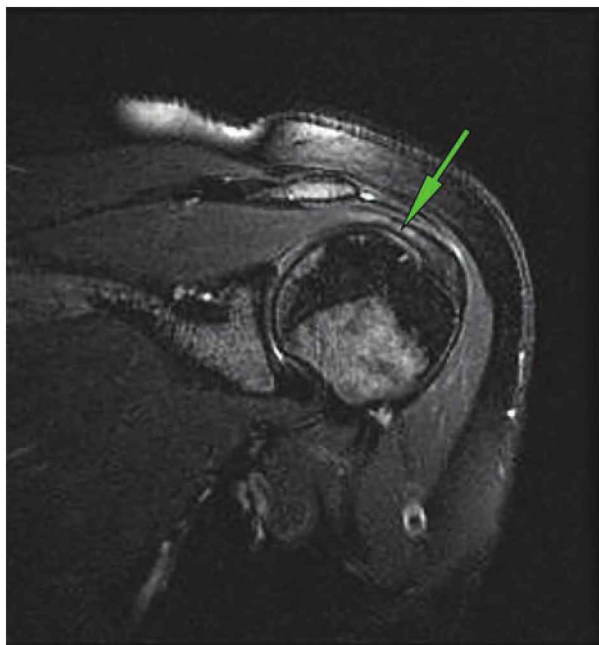


Figure 24.19 MR axial view showing supraspinatus tendinopathy (green arrow), with non-focal T2 signal increase not equal to that of fluid

Clinical dogma has suggested a corticosteroid injection into the subacromial space may reduce a patient's symptoms and allow him or her to begin shoulder rehabilitation. However, randomised trials have failed to support this position. The body of evidence in sports medicine is that corticosteroid injection provides short-term pain relief but impairs long-term recovery for tendinopathies⁵⁵ (Chapter 17).

In choosing a management plan, clinicians should consider patient preference, availability of practitioners and other health care use. If a patient prefers a more active or self-management approach, manual physical therapy, exercise and referral to a physiotherapist should be discussed. There is systematic review evidence for the benefits of exercise therapy.⁵⁶

The second part of treatment should consist of rehabilitation exercises of the rotator cuff itself. Several kinds of exercises can be applied, the selection depends upon the irritability of the tendons, the goal of the program, and the specific sport the athlete is involved in.

- Strengthening the rotator cuff (Fig. 24.20): based on EMG studies the following exercises are advised to strengthen the rotator cuff:^{57, 58} elevation in the scapular plane (supraspinatus, infraspinatus and subscapularis) (Fig. 24.20a), external rotation (supraspinatus, infraspinatus and teres minor) (Fig. 24.20b), internal rotation (subscapularis) (Fig. 24.20c), horizontal abduction

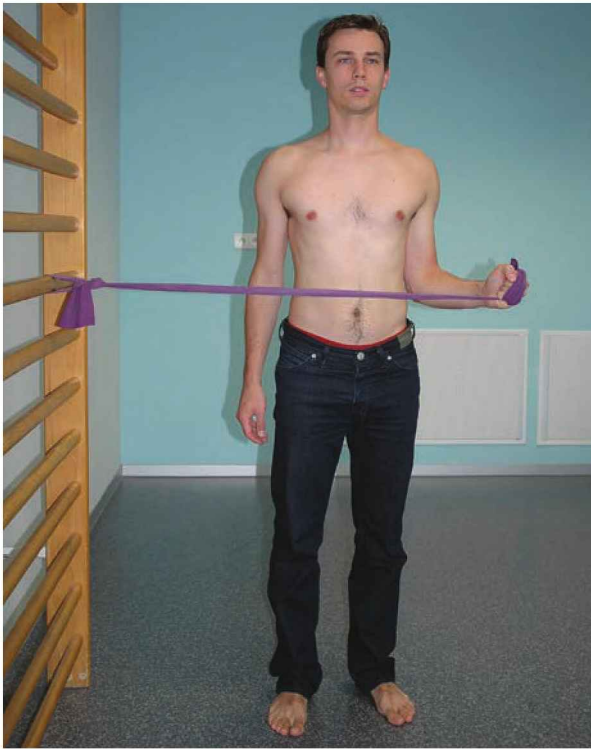
with external rotation (infraspinatus) (Fig. 24.20d). In overhead athletes special attention needs to go to the external rotators, since weakness of this muscle group is often present as a sport-specific adaptation.

- Eccentric training of the rotator cuff: evidence on structural changes in the rotator cuff tendons after eccentric training is currently lacking but clinical improvement was established.^{59, 60} Moreover, it needs to be recognised that the rotator cuff functions very much in an eccentric mode during sports activity, which strengthens the argument to include eccentric training in rehabilitation of the overhead athlete with rotator cuff tendinopathy. Exercises may be performed in a controlled slow manner, avoiding the concentric phase to allow maximal loading of the eccentric phase (Fig. 24.21), in a more plyometric way (Fig. 24.22), or accentuating the stretch-shortening cycle (Fig. 24.23).



(a)

Figure 24.20 Exercises for the rotator cuff (a) Elevation in the scapular plane



(b)



(c)

Figure 24.20 (cont.) (b) External rotation against resistance
(c) Internal rotation against resistance



(d)

Figure 24.20 (cont.) (d) Horizontal abduction with external rotation



(a)



(b)



(c)

Figure 24.21 Eccentric exercise for the external rotators
(a) Starting position (b) Increasing load (c) Ending position

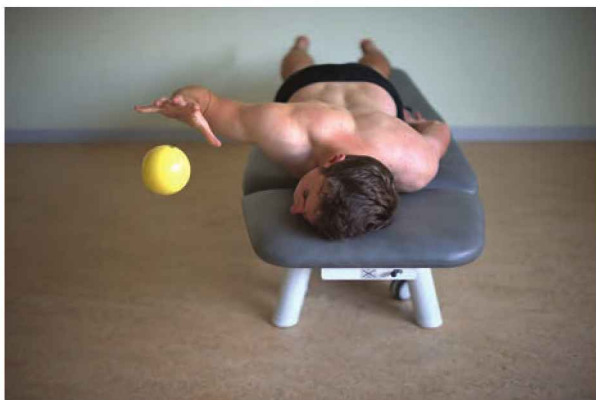


Figure 24.22 Plyometric catching exercise



Figure 24.23 XCO-Trainer® exercise

Management of calcific tendinopathy can be difficult. In general the presence of calcifications in the rotator cuff is not always associated with symptoms of impingement. The aetiology is unclear, and seems to be rather a biochemical intratendinous process in particular in the supraspinatus tendon. Calcifications are not related to rotator cuff tears or glenohumeral arthritis. In some cases ESWT (extracorporeal shock wave therapy) may be indicated, supported by evidence; however, evidence regarding the effects in non-calcific rotator cuff disease is inconclusive.⁶¹ In some cases, chronic symptomatic calcific tendinopathy is treated arthroscopically.

Rotator cuff tears

Complete and partial tears of the rotator cuff tendon are commonly seen in older athletes who present with shoulder pain during activity. These patients often complain of an inability to sleep on the affected shoulder. Examination reveals positive impingement signs and sometimes weakness on supraspinatus testing. Diagnosis is confirmed on MRI (Fig. 24.24) or ultrasound.

If the tear is small and of partial thickness, treatment may be conservative. Full thickness rotator cuff tears in young athletes require surgical repair; however, athletic performance often does not return to preoperative levels.⁶²

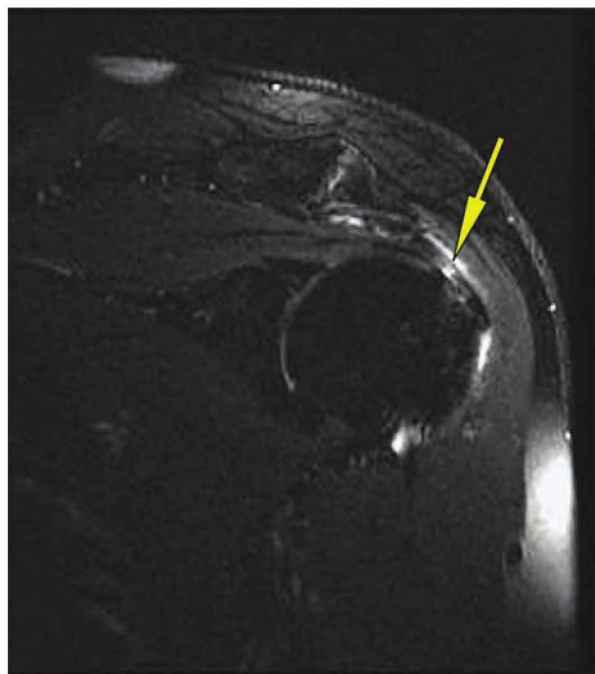


Figure 24.24 MR axial view showing partial thickness intra-substance supraspinatus tear (yellow arrow), with focal T2 signal equal to that of fluid

In older patients, still active in overhead sports, the choice for surgery or conservative treatment depends upon the degree of discomfort and pain during daily and sports activities as a result of the shoulder pain, and the expectations of the patient. Given the increasing age of people performing (overhead) sports at a recreational but also at a senior competitive level, the management of rotator cuff tears including return to sports in older athletes gains interest. Analysing the definition of a master athlete as: 'Active individual aged 50 years or older, who desires optimal levels of performance or wishes to exercise for general health and has high expectations for sports medicine care, including return to sport or activity after injury', achieving these goals is a huge challenge for all healthcare professionals involved with older athletes.

Although there is no general consensus on the approach of rotator cuff tears, there seems to be a shift towards conservative treatment.^{63, 64} Although not proven to be efficient, we advise to perform exercises with minimal load on the rotator cuff, and maximally aim to restore function, in order to postpone the 'final match'. Exercises improving function (elevation) with minimal load on the rotator cuff are:

- semi-closed chain exercises with increasing gravity impact and increasing resistance⁶⁵ (Fig. 24.25)
- shoulder balance exercises in increasing elevation⁶⁶ (Fig. 24.26).

SHOULDER INSTABILITY

Based on the cause, direction and typical clinical presentation of instability, patients may be divided into three groups: the TUBS (traumatic unidirectional instability with Bankart lesion, for which surgery is often needed), AIOS (acquired instability due to overstress syndrome), and AMBRI (atraumatic multidirectional instability with bilateral laxity, in which rehabilitation is mandatory, but in case of failure inferior capsular shift surgery is performed). On the field these types may be combined, for instance a gymnast with general laxity (AMBRI) may develop overuse instability (AIOS), or a volleyball player with sports-specific minor instability (AIOS) may experience an acute trauma and dislocation (TUBS).

Traumatic shoulder instability—TUBS

One of the most common traumatic sports injuries is acute dislocation of the glenohumeral joint (Fig. 24.27a, b). In almost all cases, this is an anterior dislocation and it results from the arm being forced into excessive abduction and external rotation. Most anterior dislocations damage the attachment of the labrum to the anterior glenoid margin (Bankart lesion) as shown on MRI (Fig. 24.27c).



(a)



(b)

Figure 24.25 Semi-closed chain exercises against external resistance (a) Without gravity (b) Against gravity



Figure 24.26 Shoulder balance exercise